AFZAL AHMAD

https://afxalz.github.io/
afzalhmd14@gmail.com

EDUCATION

Czech Technical University in Prague, Czech Republic 2020 - 2022
Master of Science (M.Sc) in Computer Science
Department of Computer Science

Indian Institute of Technology (IIT) Guwahati, India 2015 - 2019
Bachelor of Technology (B.Tech) in Electrical Engineering
Department of Electronics and Electrical Engineering

PUBLICATIONS

Tatana Prihodova, Giuseppe Silano, **Afzal Ahmad**, Vit Kratky, Tomas Baca, Pavel Petracek, Vera Saskova, Jan Bednar and Martin Saska.*"2022 IEEE Robotics and Automation Society Summer School on Multi-Robot Systems in Prague [Education]"*, IEEE Robotics & Automation Magazine (RAM), March 2023.

Afzal Ahmad, Daniel Bonilla Licea, Giuseppe Silano, Tomas Baca and Martin Saska. *"PACNav: A Collective Navigation Approach for UAV Swarms Deprived of Communication and External Localization"*, Bioinspiration & Biomimetics journal (BnB), Nov. 2022.

Afzal Ahmad, Viktor Walter, Pavel Petracek, Matej Petrlik, Tomas Baca, David Zaitlik and Martin Saska.*"Autonomous Aerial Swarming in GNSS-denied Environments with High Obstacle Density"*, IEEE International Conference on Robotics and Automation (ICRA) 2021.

Andriy Dmytruk, Tiago Nascimento, **Afzal Ahmad**, Tomas Baca, and Martin Saska. *"Safe Tightly-Constrained UAV Swarming in GNSS-denied Environments"*, International Conference on Unmanned Aircraft Systems (ICUAS) 2021.

Afzal Ahmad, Vojtech Vonasek and Martin Saska. *"Cooperative path planning for multiple MAVs operating in unknown environments"*, International Conference on Unmanned Aircraft Systems (ICUAS) 2020.

EXPERIENCE

Research Engineer Dec 2022 - June 2025
Fly4Future, Prague, Czech Republic

- Worked on creating client-facing multi-robot systems for applications including fire-fighting, outdoor inspection and search-and-rescue missions.
- Focused on control and motion planning algorithms to enable a group of robots to work closely with human operators using web-UI or Command Line interfaces.
- Designed different testing pipelines using ROS2 and Gazebo for software-in-the-loop and hardware-in-the-loop analysis of the integrated system.
- Successfully deployed the mission execution software stack on remote hardware using Docker/Singularity.

Graduate Research Fellow Aug 2019 - Dec 2022
Multi-Robot Systems Lab, Prague, Czech Republic

- Designed one of the first aerial robotic swarm to operate in GPS-denied environments. The UAVs could operate without any communication with others. The work was featured in the IEEE Spectrum magazine and published in **ICRA 21**.

- Developed a aerial robotic swarming algorithm using observed trajectories of neighboring UAVs to ensure collective motion in forest-like environments which was published in a **journal**.
- Worked extensively with the open-source MRS-UAV system for rapid prototyping and deployment of multi-robot systems in simulated and field experiments.

Research Intern May 2018 - July 2018
MIST Lab, Polytechnique Montreal, Canada

- Worked on a ROS-based architecture for path planning and control of multi-UAV systems.
- Developed a framework for long-term autonomy of a multi-robot system using adaptive locomotion.

Research Intern June 2017 - Dec 2017
Indian Institute of Technology (IIT) Guwahati, India

- Developed a model-based learning technique for locomotion of a nonholonomic snake robot (ASROA).
- Used the learned model to implement obstacle avoidance in a simulated environment.

TECHNICAL SKILLS

Programming Languages	C++, Python, C, Bash
Softwares & Tools	Gazebo, Docker, Singularity, Git, LaTeX, Cmake
Operating Systems	Ubuntu, Robot Operating System (ROS1 and ROS2)
Hardware	Intel NUC, Raspberry Pi, Pixhawk Autopilot, Arduino

HONORS AND AWARDS

- Selected among the **27** fellows for the **Robotics: Science and Systems** inclusion fellowship.
- Selected among the **top 6** master's thesis dissertations for the **Cena Wernera von Siemense awards** under the **Industry 4.0** category (for potential impact on the new age of industrial revolution).
- Awarded a scholarship to cover tuition fees for the master's program at **Czech Technical University in Prague**.
- Awarded the **Shastri Research Student Fellowship** to pursue a research internship at **Polytechnique Montreal, Canada** (only awarded to 18 students all across India).

EDITORIAL ACTIVITIES

- Reviewed articles for IEEE RA-L and Bioinspiration Biomimetics journals.
- Reviewed articles for IROS and ICRA conferences.